# Imitation Learning

## CISC 7404 - Decision Making

Steven Morad

University of Macau

Admin . . . . . 2
Final Projects . . . . . 3
Finish DPG . . . . . 4
Imitation Learning . . . . . 5
Behavioral Cloning . . . . . 7
Coding . . . . . 8
Applications . . . . . 10

# Admin

# Admin

TAs currently grading homework 1:

- Grades will be released over the next week on Moodle
- All grades should be released by Wednesday

Homework 2 should be graded by the following Wednesday (April 16)

# Admin

I am currently grading quiz 2, 40% done, score released soon

- I want scoring to be consistent for all students
  - I grade exams myself (not TA), so grading is slow

Highly multimodal distribution

- Modes at 20%, 45%, and 80%
  - I drop lowest quiz score, if you did poorly study more next time
- Some students received perfect scores, good job!
  - He Enhao
  - Leonard Hangqin Zhuang
  - Qiao Yulin
  - Fu Zexin

# Admin

We must have one more quiz

Current options:

- 17 April
- 24 April (last lecture)

Leaning towards 17 April so we can spend all of 24 on LLMs

# Final Projects

# Final Projects

Project plans turned in

- Legend of Zelda
- Super Mario (7x)
- LLM finetuning
- Honor of Kings (3x)
- Federated learning
- Sudoku
- Snake (2x)
- Health AI system
- Tetris
- PacMan (2x)
- Navigation
- StarCraft II
- Getting Over It
- Gymnax
- 2048
- Pokemon Double Battle

# Final Projects

Most plans look good

I left some comments/suggestions in Moodle

I want to highlight the most creative project plan (in my opinion)

*Getting Over It by Bennet Foddy* Group: Getting Over It

I asked for projects that can improve the world

No human should ever have to play *Getting Over It*

Training an agent to play instead can improve many human lives

At university, my friend skipped school for a week to win this game

https://youtu.be/8qGCleYV4cw?si=ynB0Idg5-TdAiAh9&t=74

# Final Projects

Dr. Bennet Foddy (game author) teaches at NYU

His research focus is on addiction and reward/punishment

> This chapter compares … data on different addictions … from drug addiction to binge-eating disorders, gambling, and videogame addiction. … Based on these data, it is argued that **there is a hazard inherent in any rewarding operant behavior, no matter how apparently benign: that we may become genuinely "addicted" to any behavior that provides operant reward**. With this in mind, addiction is rightly seen as a possibility for any human being, not a product of the particular pharmacological or technological properties of any one particular substance or behavior.

# Final Projects

We become addicted (gambling, drugs, etc) because of reward!

Drugs/gambling/etc reward is so powerful it overwhelms the rewards of normal life

$$\text{study} + \gamma \text{ exercise} + \gamma^2 \text{ sleep} + \ldots = 10$$

$$\text{no money} + \gamma \text{ no sleep} + \gamma^2 \text{ pain} + \ldots + \underbrace{\gamma^n \text{ addiction pleasure}}_{\text{Too powerful}} = 100$$

If you have no addiction, then you are not following an optimal policy

Optimal policy in humans **will** lead to bad behavior

- Addict behavior policy maximizes the return!

# Final Projects

*Getting Over It* represents this addiction

$$\text{fall} + \gamma\ \text{anger} + \ldots + \gamma^n\ \text{win game} > \text{study} + \gamma\ \text{exercise} + \ldots$$

It is always interesting to consider our own reward functions

Humans are born with reward functions and cannot choose them

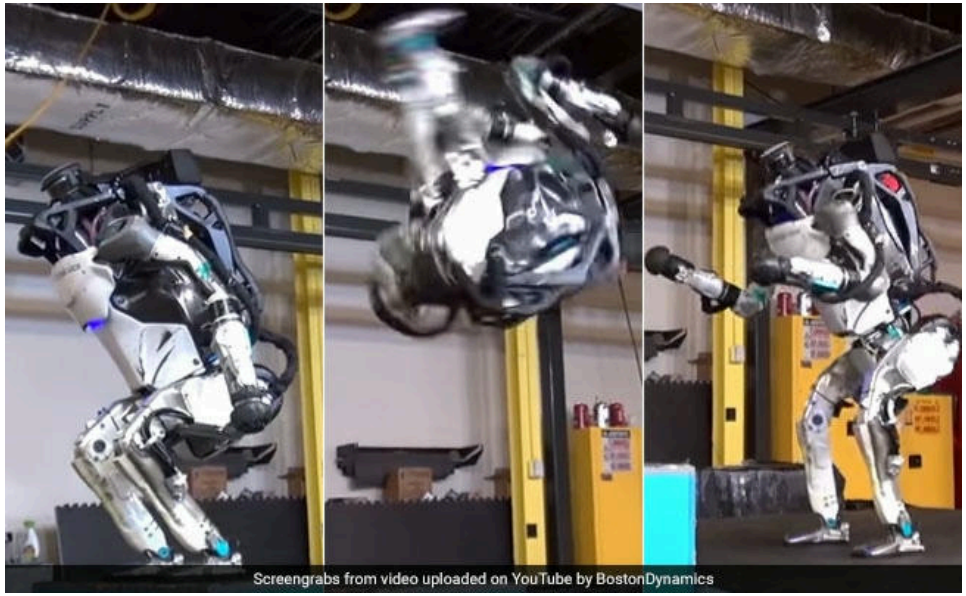
We can select good reward functions for artificial agents

LLMs are patient, intelligent, helpful because of the reward function

# Finish DPG

# Imitation Learning

# Imitation Learning




**Example:** You are a scientist at Boston Dynamics/Tencent Robotics/etc

You need to release demos to impress the public

You want to learn a backflip policy using RL

**Question:** What reward function should you use?

$$\mathcal{R}(s) =$$

# Imitation Learning

Policy will maximize the return, but not exhibit desired behavior

Writing reward functions can be difficult

Rewards are particularly difficult when interacting with humans

We want our robot to:

- Be friendly
- Be polite
- Not scare humans

It is very hard to write a reward function for these behaviors

# Imitation Learning

It is often easier to demonstrate good behavior than create rewards

How would a human do a backflip?

Jump

Backward rotation

$$\pi(a_0 \mid s_0, \theta_{\text{human}}), \pi(a_1 \mid s_1, \theta_{\text{human}}), \pi(a_2 \mid s_2, \theta_{\text{human}}) \ldots$$

Tuck legs

In **imitation learning** we learn to imitate an expert policy

I want to introduce a formalism to model the problem

# Imitation Learning

**Definition:** A Hidden-Reward MDP (HR-MDP) consists of:

- MDP with a **hidden** reward function
- Dataset $\boldsymbol{X}$ from following an expert policy $\pi\big(a \mid s; \theta_\beta\big)$

$$(S, A, \mathcal{R}, \mathrm{Tr}, \gamma, \boldsymbol{X})$$

$$s_{t+1} \sim \mathrm{Tr}(\cdot \mid s_t, a_t)$$

$$\mathcal{R}(s_{t+1}) = ?$$

$$\boldsymbol{X} = [\boldsymbol{\tau}_1 \;\; \boldsymbol{\tau}_2 \;\; \ldots] = \begin{bmatrix} \begin{bmatrix} (s_0, a_0) \\ (s_1, a_1) \\ \vdots \end{bmatrix} & \begin{bmatrix} (s_0, a_0) \\ (s_1, a_1) \\ \vdots \end{bmatrix} & \ldots \end{bmatrix}$$

# Imitation Learning

https://www.youtube.com/watch?v=4N4czAm61Fc

# Behavioral Cloning

# Behavioral Cloning

Expert policy $\pi(a \mid s; \theta_\beta)$

**Question:** What is our objective?

$$\pi(a \mid s; \theta_\pi) = \pi(a \mid s; \theta_\beta)$$

$$\arg\min_{\theta_\pi} \left(\pi(a \mid s; \theta_\beta) - \pi(a \mid s; \theta_\pi)\right)^2$$

**Question:** Does this work?

**Answer:** No, $\pi(a \mid s)$ is a distribution not a scalar

**Question:** How can we measure the difference of two distributions?

**Answer:** KL divergence measures difference in distributions

# Behavioral Cloning

$$\mathrm{KL}(\Pr(X), \Pr(Y)) = \sum_{\omega \in \Omega_X} \Pr(X = \omega) \log \frac{\Pr(X = \omega)}{\Pr(Y = \omega)}$$

Minimize difference between $\underbrace{\pi\big(a \mid s; \theta_\beta\big)}_{\Pr(X)}$ and $\underbrace{\pi(a \mid s; \theta_\pi)}_{\Pr(Y)}$

$$\arg\min_{\theta_\pi} \ \mathrm{KL}\big(\pi\big(a \mid s; \theta_\beta\big), \pi(a \mid s; \theta_\pi)\big)$$

Plug policies into KL equation

$$\arg\min_{\theta_\pi} \sum_{a \in A} \pi\big(a \mid s; \theta_\beta\big) \log \frac{\pi\big(a \mid s; \theta_\beta\big)}{\pi(a \mid s; \theta_\pi)}$$

# Behavioral Cloning

$$\arg\min_{\theta_\pi} \sum_{a \in A} \pi(a \mid s; \theta_\beta) \log \frac{\pi(a \mid s; \theta_\beta)}{\pi(a \mid s; \theta_\pi)}$$

Log of divisors is difference of logs

$$\arg\min_{\theta_\pi} \sum_{a \in A} \pi(a \mid s; \theta_\beta) \left[\log(\pi(a \mid s; \theta_\beta)) - \log(\pi(a \mid s; \theta_\pi))\right]$$

Distribute behavior policy into difference

$$\arg\min_{\theta_\pi} \sum_{a \in A} \pi(a \mid s; \theta_\beta) \log \pi(a \mid s; \theta_\beta) - \pi(a \mid s; \theta_\beta) \log \pi(a \mid s; \theta_\pi)$$

# Behavioral Cloning

$$\arg\min_{\theta_\pi} \sum_{a \in A} \pi(a \mid s; \theta_\beta) \log \pi(a \mid s; \theta_\beta) - \pi(a \mid s; \theta_\beta) \log \pi(a \mid s; \theta_\pi)$$

First term is constant with respect to $\theta_\pi$ (only depends on $\theta_\beta$)

Can ignore the first term for optimization purposes

$$\arg\min_{\theta_\pi} \sum_{a \in A} -\pi(a \mid s; \theta_\beta) \log \pi(a \mid s; \theta_\pi)$$

This is the **cross-entropy** objective

**Question:** Have we seen this objective in deep learning?

**Answer:** Loss for classification tasks

# Behavioral Cloning

$$\arg\min_{\theta_\pi} \sum_{a \in A} -\pi(a \mid s; \theta_\beta) \log \pi(a \mid s; \theta_\pi)$$

**Question:** Where does $s$ come from? **Answer:** Dataset $\boldsymbol{X}$

$$\arg\min_{\theta_\pi} \sum_{s \in \boldsymbol{X}} \sum_{a \in A} -\pi(a \mid s; \theta_\beta) \log \pi(a \mid s; \theta_\pi)$$

# Behavioral Cloning

**Definition:** Behavioral cloning learns the parameters $\theta_\pi$ to match an expert policy $\theta_\beta$ that collected the dataset $\boldsymbol{X}$

$$\arg\min_{\theta_\pi} \sum_{s \in \boldsymbol{X}} \sum_{a \in A} -\pi\big(a \mid s; \theta_\beta\big) \log \pi(a \mid s; \theta_\pi)$$

Behavioral cloning is a simple supervised learning algorithm!

# Behavioral Cloning

$$\arg\min_{\theta_\pi} \sum_{s \in \boldsymbol{X}} \sum_{a \in A} -\pi\big(a \mid s; \theta_\beta\big) \log \pi(a \mid s; \theta_\pi)$$

**Question:** How do we evaluate this?

If $A$ is discrete, we can sum over possible actions $a \in A$

- Our expert policy will often be one-hot
- $\pi\big(a = a_* \mid s; \theta_\beta\big) = 1$ (action the expert took)
- $\pi\big(a \neq a_* \mid s; \theta_\beta\big) = 0$ (all other actions)

What if $A$ is continuous (infinite)?

# Behavioral Cloning

$$\arg\min_{\theta_\pi} \sum_{s \in \boldsymbol{X}} \sum_{a \in A} -\pi(a \mid s; \theta_\beta) \log \pi(a \mid s; \theta_\pi)$$

If $A$ is continuous, we have an infinite sum

$$\arg\min_{\theta_\pi} \sum_{s \in \boldsymbol{X}} \int_A -\pi(a \mid s; \theta_\beta) \log \pi(a \mid s; \theta_\pi) \, \mathrm{d}a$$

Not all integrals are solvable

Need to be careful how we model $\pi$, to make sure we can solve this

# Behavioral Cloning

$$\arg\min_{\theta_\pi} \sum_{s \in \boldsymbol{X}} \int_A -\pi\big(a \mid s; \theta_\beta\big) \log \pi(a \mid s; \theta_\pi)\, \mathrm{d}a$$

To ensure we have an analytical solution to this integral:

- Use Dirac delta expert policy $\pi\big(a \mid s; \theta_\beta\big) = \delta(a - a_*)$
- Use Gaussian learned policy $\pi(a \mid s; \theta_\pi) = \mathrm{Normal}(\mu, \sigma)$

# Behavioral Cloning

$$\arg\min_{\theta_\pi} \sum_{s \in \boldsymbol{X}} \int_A -\pi(a \mid s; \theta_\beta) \log \pi(a \mid s; \theta_\pi) \, \mathrm{d}a$$

To ensure we have an analytical solution to this integral:

- Use Dirac delta expert policy $\pi(a \mid s; \theta_\beta) = \delta(a - a_*)$
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Plug our policies into the cross entropy objective

Expert action taken

$$\arg\min_{\theta_\pi} \sum_{s \in \boldsymbol{X}} \int_A -\underbrace{\delta(a - a_*)}_{\text{Expert policy}} \log \pi(a \mid s; \theta_\pi) \, \mathrm{d}a$$

# Behavioral Cloning

$$\arg\min_{\theta_\pi} \sum_{s \in \boldsymbol{X}} \int_A -\underbrace{\delta(a - a_*)}_{\text{Expert policy}} \log \pi(a \mid s; \theta_\pi) \, \mathrm{d}a$$

**Question:** What happens to $\int_A \delta(a - a_*) f(a) \, \mathrm{d}a$?

**Answer:** Becomes $f(a = a_*)$

$$\arg\min_{\theta_\pi} \sum_{s \in \boldsymbol{X}} \cancel{\int_A -\delta(a - a_*) \log \pi(a \mid s; \theta_\pi) \mathrm{d}a}$$

$$\arg\min_{\theta_\pi} \sum_{s \in \boldsymbol{X}} -\log \pi(a = a_* \mid s; \theta_\pi)$$

# Behavioral Cloning

$$\arg\min_{\theta_\pi} \sum_{s \in \boldsymbol{X}} -\log \pi(a = a_* \mid s; \theta_\pi)$$

Now, plug in Gaussian distribution for learned policy

$$\pi(a = a_* \mid s; \theta_\pi) = \text{Normal}(a = a_* \mid \mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(a_* - \mu)^2}{2\sigma^2}\right)$$

$$\arg\min_{\theta_\pi} \sum_{s \in \boldsymbol{X}} -\log \left[\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(a_* - \mu)^2}{2\sigma^2}\right)\right]$$

# Behavioral Cloning

$$\arg\min_{\theta_\pi} \sum_{s \in \boldsymbol{X}} -\log\left[\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(a_* - \mu)^2}{2\sigma^2}\right)\right]$$

Log of products is sum of logs

$$\arg\min_{\theta_\pi} \sum_{s \in \boldsymbol{X}} -\log\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right) - \log\left(\exp\left(-\frac{(a_* - \mu)^2}{2\sigma^2}\right)\right)$$

Log of divisors is difference of logs

$$\arg\min_{\theta_\pi} \sum_{s \in \boldsymbol{X}} -\log(1) + \log\left(\sqrt{2\pi\sigma^2}\right) - \log\left(\exp\left(-\frac{(a_* - \mu)^2}{2\sigma^2}\right)\right)$$

# Behavioral Cloning

$$\arg\min_{\theta_\pi} \sum_{s \in \boldsymbol{X}} -\log(1) + \log\left(\sqrt{2\pi\sigma^2}\right) - \log\left(\exp\left(-\frac{(a_* - \mu)^2}{2\sigma^2}\right)\right)$$

$$\log(1) = 0$$

$$\arg\min_{\theta_\pi} \sum_{s \in \boldsymbol{X}} \log\left(\sqrt{2\pi\sigma^2}\right) - \log\left(\exp\left(-\frac{(a_* - \mu)^2}{2\sigma^2}\right)\right)$$

Rewrite $\sqrt{\ldots}$ as power

$$\arg\min_{\theta_\pi} \sum_{s \in \boldsymbol{X}} \log\left(\left(2\pi\sigma^2\right)^{\frac{1}{2}}\right) - \log\left(\exp\left(-\frac{(a_* - \mu)^2}{2\sigma^2}\right)\right)$$

# Behavioral Cloning

$$\arg\min_{\theta_\pi} \sum_{s \in \boldsymbol{X}} \log\left(\left(2\pi\sigma^2\right)^{\frac{1}{2}}\right) - \log\left(\exp\left(-\frac{\left(a_* - \mu\right)^2}{2\sigma^2}\right)\right)$$

Log of power is product of power and log

$$\arg\min_{\theta_\pi} \sum_{s \in \boldsymbol{X}} \frac{1}{2}\log(2\pi\sigma^2) - \log\left(\exp\left(-\frac{\left(a_* - \mu\right)^2}{2\sigma^2}\right)\right)$$

Now simplify the second term, log and exp cancel

$$\arg\min_{\theta_\pi} \sum_{s \in \boldsymbol{X}} \frac{1}{2}\log(2\pi\sigma^2) + \frac{\left(a_* - \mu\right)^2}{2\sigma^2}$$

# Behavioral Cloning

$$\arg\min_{\theta_\pi} \sum_{s \in \boldsymbol{X}} \frac{1}{2} \log(2\pi\sigma^2) + \frac{(a_* - \mu)^2}{2\sigma^2}$$

Since this is an optimization problem, we can drop the constant $2\pi$

$$\arg\min_{\theta_\pi} \sum_{s \in \boldsymbol{X}} \frac{1}{2} \log \sigma^2 + \frac{(a_* - \mu)^2}{2\sigma^2}$$

Clean up a bit by factoring

$$\arg\min_{\theta_\pi} \sum_{s \in \boldsymbol{X}} \frac{1}{2} \left( \log \sigma^2 + \frac{(a_* - \mu)^2}{\sigma^2} \right)$$

# Behavioral Cloning

For a Gaussian policy, we minimize the following objective

$$\arg\min_{\theta_\pi} \sum_{s \in \boldsymbol{X}} \frac{1}{2} \left( \log \sigma^2 + \frac{(a_* - \mu)^2}{\sigma^2} \right)$$

**Note:** We derive the loss function for a Gaussian policy gradient loss the exact same way

# Behavioral Cloning

**Definition:** Behavioral cloning uses **supervised learning** for decision making, minimizing the cross-entropy between expert $\theta_\beta$ and our $\theta_\pi$

$$\arg\min_{\theta_\pi} \sum_{s,a \in \boldsymbol{X}} -\pi(a \mid s; \theta_\beta) \log \pi(a \mid s; \theta_\pi)$$

For continuous actions, we pick special distributions so the cross entropy loss is tractable (Dirac delta $\theta_\beta$, Gaussian $\theta_\pi$)

Expert action

$$\arg\min_{\theta_\pi} \sum_{s \in \boldsymbol{X}} \frac{1}{2} \left( \log \sigma^2 + \frac{(a_* - \mu)^2}{\sigma^2} \right)$$

Policy outputs

# Coding

# Coding

Similar to policy gradient, just with different loss function

The policies and methods change depending on action space

Discrete actions use a categorical distribution

Continuous actions usually use normal distribution

Start with discrete actions, then do continous

# Coding

Implement a model for a categorical policy

```
model = Sequential([
    Linear(state_size, hidden_size),
    Lambda(leaky_relu),
    Linear(hidden_size, hidden_size),
    Lambda(leaky_relu),
    # Output logits (real numbers)
    Linear(hidden_size, action_size),
])
```

Identical to policy gradient

# Coding

Next, implement discrete loss function

```
def bc_loss(model, states, actions):
    # Often, we can't know the expert action distribution
    # We only have the taken expert action
    # Taken action has p=1, all other actions p=0
    # Represent as a one-hot vector
    expert_probs = actions
    log_policy_probs = log_softmax(vmap(model)(states))
    # Log loss, can reduce over batch using mean or sum
    bce_loss = -sum(
        expert_probs * log_policy_probs, axis=1).mean()
    return bce_loss
```

# Coding

Finally, to run our policy we sample actions from our policy

```
def sample_action(model, state, key):
    z = model(state)
    # BE VERY CAREFUL, always read documentation
    # Sometimes takes UNNORMALIZED logits, sometimes probs
    action_probs = softmax(model, state)
    a = categorical(key, action_probs)
    a = categorical(key, z) # Does not even use pi
    return a
```

Identical to policy gradient

# Coding

Now, consider Gaussian policy (continuous actions)

```
model = Sequential([
    Linear(state_size, hidden_size),
    Lambda(leaky_relu),
    Linear(hidden_size, hidden_size),
    Lambda(leaky_relu),
    # Output mu and log_sigma
    Linear(hidden_size, 2 * action_size),
    Lambda(x: split(x, 2))
])
```

Same as policy gradient

# Coding

Implement continuous loss function

Use simplified cross entropy (Dirac-Gaussian)

```
def bc_loss(model, states, actions):
    expert_probs = actions # Dirac delta
    mu, log_std = vmap(model)(states)
    # Gaussian CE, also called Gaus. Neg. Log Likelihood
    gnll_loss = log_std + 0.5 * (
        (mu - action)**2 / exp(log_std)**2
    )
    return gnll_loss
```

# Coding

Next, we need to sample actions from our policy network

```
def sample_action(model, state, key):
    mu, log_sigma = model(state)
    # Reparameterization trick
    noise = random.normal(key, (action_size,))
    a = mu + exp(log_sigma) * noise
    return a
```

# Coding

The training loop is much simpler than RL

```
model = Sequential(...)
opt_state = ...
# Just supervised learning
for batch in dataset:
    states, actions = batch
    J = grad(bc_loss)(model, states, actions)
    update = optim.update(J, opt_state)
    model = apply_updates(update, model)
```

Then deploy the policy

```
a_t = sample_action(model, s_t, key)
```

# Applications

# Applications

Behavioral cloning seems very powerful!

- No need for reward function
- No need for exploration, learn from fixed dataset
- Simpler to implement than RL
- More stable to train than RL

Sounds very promising, why do we care about RL?

**Question:** What are some disadvantages of BC?

# Applications

**Limitation:** Imperfect expert

Dataset is following an "expert" $\theta_\beta$

Humans are not reliable experts in many cases

You will learn a policy that drives like it is texting

# Applications

Even where all the data is from a reliable "expert", we have problems

Consider a human surgeon

Human surgeons are not as fast or precise as machines

Even the best machine will be no better than a human surgeon

$$\pi(a \mid s; \theta_\pi) \approx \underbrace{\pi\big(a \mid s; \theta_\beta\big)}_{\text{Human policy}}$$

**Question:** Any other issues?

# Applications

**Limitation:** Humans limited to small regions of state space

Humans are very bad at exploring

There will be very few crashes in a self driving dataset

When the policy is about to crash, the state will be out of distribution

The policy will be unable to avoid crashing

In practice, BC policies generalizes much worse than RL policies

Small errors in the learned policy eventually drive the policy to out of distribution states

# Applications

One solution to out of distribution error is to collect more data

**DAgger:** The agent asks an expert for help when it visit an out of distribution state

- This requires an expert to always be paying attention

**Inverse RL:** We learn the reward function that the expert is maximizing

$$\arg\max_{\theta_R} \mathbb{E}\left[\mathcal{G}(\boldsymbol{\tau}) \mid s_0; \theta_\beta\right]$$

$$= \arg\max_{\theta_R} \sum_{n=0}^{\infty} \gamma^n \sum_{s_{n+1} \in S} \underbrace{\mathcal{R}(s_{n+1}, \theta_R)}_{\text{Learnable}} \cdot \Pr\left(s_{n+1} \mid s_0; \theta_\beta\right)$$

Then, we learn a policy $\theta_\pi$ using RL. This generalizes better than BC